IMO(3^{rd}) – 1961

First Day

1. Solve the following system of equations: $x+y+z=a,\ x^2+y^2+z^2=b^2,\ xy=z^2$, where a,b are given real numbers. What conditions must hold on a,b for the solutions to be positive and distinct?
2. Let a,b,c be the lengths of a triangle whose area is S. Prove that $a^2+b^2+c^2 \geq 4S\sqrt{3}$. In what case does equality hold?
3. Solve the equation $\cos^n x - \sin^n x = 1$, where n is a given positive integer.

Second Day

4. In the interior of $\Delta P_1P_2P_3$ a point P is given. Let Q_1, Q_2, Q_3 respectively be the intersection of PP_1, PP_2, PP_3 with the opposite edges of $\Delta P_1P_2P_3$. Prove that among the ratios $PP_1/PQ_1,\ PP_2/PQ_2,\ PP_3/PQ_3$ there exist at least one not larger that 2 and at least one not smaller than 2.
5. Construct a triangle ABC if the following elements are given: $AC=b, AB=c,$ and $\angle AMB = w (w < 90^0)$, where M is the mid point of BC. Prove that the construction has a solution if and only if $b\tan\frac{w}{2} \leq c < b$. In what case does equality hold?
6. A plane ε is given and on one side of the plane three non collinear points A, B, and C such that the plane determined by them is not parallel to ε. Three arbitrary A', B', C' in ε are selected. Let L, M and N be the mid points of AA', BB', CC' and G the centroid of ΔLMN. Find the locus of all points obtained for G as A', B', C' are varied (independently of each other) across ε.

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